Lightweight AI-Based Small Drone Detection, Visual Tracking, and Short-Term Trajectory Prediction Under Occlusion and Camera Motion

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***Abstract*— Low-altitude micro-unmanned aerial vehicles (UAVs) are difficult to monitor reliably with onboard or ground-based cameras when computational power is limited. In real-world flights, four problems occur together: distant quadrotors subtend fewer than 24×24 pixels, urban buildings and tree foliage lower detector confidence, obstacles briefly block the line of sight, and camera pan or tilt motion shifts bounding boxes across the image plane. In this paper, we develop and test a lightweight aerial tracking pipeline that couples a modified YOLOv8n-P2 small-target detector, two-stage ByteTrack data association, monocular Visual-Inertial Odometry (VIO), and a 3D Constant-Turn-Rate and Acceleration Extended Kalman Filter (CTRA-EKF). A high-resolution 160×160 P2 feature head with Wise-IoU regression captures sub-20-pixel drones, while tightly coupled VIO fuses 30 Hz visual features with 200 Hz inertial measurement unit (IMU) pre-integration to compensate camera ego-rotation and translation prior to ByteTrack association. During complete visual occlusions (1.0–3.0 s), the 3D CTRA-EKF propagates curvilinear target dynamics to prevent track fragmentation. Evaluated across 16 physics-based aerial surveillance sequences (14,400 frames) in Gazebo Harmonic 8.15 alongside standard UAV small-object benchmarks, our framework achieves 91.4% AP_{50}, 89.4% MOTA, 91.2% IDF1, and 81.6% HOTA while reducing identity switches by 78.9% over uncompensated baselines. Camera ego-motion estimation reaches an Absolute Trajectory Error (ATE_{RMSE}) of 0.084 m, while the 3D CTRA-EKF lowers 2.0 s blind-occlusion prediction RMSE by 81.8% (0.36 m vs. 1.98 m). The full pipeline runs at 128.2 FPS (7.8 ms per frame) on a desktop GPU and 46.1 FPS on an embedded Jetson Orin Nano. All simulation worlds, ROS 2/Gazebo nodes, and evaluation datasets are publicly available at* `https://github.com/T-Storm-072/03_uav_drone_tracking_icect2027`.

***Index Terms*— Small drone detection, YOLOv8, ByteTrack, Extended Kalman Filter (EKF), Visual-Inertial Odometry (VIO), occlusion handling, Gazebo simulation.**

I Introduction

Small rotary-wing Unmanned Aerial Vehicles (UAVs) are now routine in crop monitoring, bridge inspection, search-and-rescue, and site security [1, 2]. At the same time, low-cost micro-quadrotors have made airspace monitoring much harder, requiring vision systems that can spot, track, and predict fast aerial trajectories on low-power hardware [3].

When a camera tracks a small drone outdoors, four failure modes interact directly. First, beyond 25 m standoff range, a 0.28 m quadrotor covers only 8×8 to 24×24 pixels ($< 0.08\%$ of a 1280×720 image), so standard stride-8 and stride-16 backbone layers lose its spatial features [1, 4]. Second, brick facades, moving branches, and changing sky brightness cause confidence scores to drop and trigger false alarms [5, 6]. Third, when a drone passes behind a pillar or tree canopy, visual measurements disappear for several seconds; standard constant-velocity Kalman filters then drift along the pre-occlusion tangent and assign a new track ID when the drone reappears [9, 10]. Fourth, if the camera itself rotates on a gimbal or flies onboard a pursuit UAV, camera pitch and yaw shifts move bounding boxes across the frame even when the target hovers still [11, 12].

While recent studies have independently advanced lightweight YOLO architectures for small aerial targets [1, 3, 7, 8], multi-stage detection association via ByteTrack [9], and monocular Visual-Inertial Odometry (VIO) via VINS-Mono [12], existing literature largely evaluates these modules in isolation. Static-camera or downward-looking ground-target benchmarks (such as standard VisDrone pedestrian/vehicle tasks [2]) do not establish how detector confidence drops, camera ego-motion drift, and multi-second blind occlusions interact during dynamic air-to-air or ground-to-air drone tracking.

In this work, we address these four failure modes together inside a unified ROS 2 and Gazebo Harmonic 8.15 simulation framework. Specifically, our contributions are:

- **High-Resolution Small-UAV Detector:** An enhanced YOLOv8n-P2 architecture incorporating a 160×160 shallow feature pyramid branch and dynamic Wise-IoU bounding-box regression tailored for sub-24-pixel drones.
- **VIO Ego-Motion Compensated ByteTrack:** Tightly coupled integration of 200 Hz IMU pre-integration and monocular visual odometry ($\mathbf{T}_w^{c_t} \in SE(3)$) to project tracklet priors across aggressive camera pan/tilt jitter prior to two-stage ByteTrack association.
- **Curvilinear 3D CTRA-EKF Occlusion Predictor:** A nonlinear Extended Kalman Filter modeling 3D target velocity, yaw angle, and turn rate to bridge 1.0–3.0 s complete visual occlusions with minimal spatial drift.
- **Comprehensive Gazebo & Benchmark Evaluation:** Rigorous ablation across 16 simulated sequences (14,400 frames) and 4 environmental regimes (R_1–R_4), quantifying accuracy (AP_{50}, MOTA, IDF1, HOTA, RMSE, ATE_{RMSE}) and real-time computational efficiency (FPS, latency, and memory).

II Literature Review

A Lightweight Small-Object Detection in Aerial Imagery

Objects occupying only a few dozen pixels suffer from spatial information loss during strided downsampling in deep convolutional networks [2]. Xu et al. [1] improved YOLOv8n on

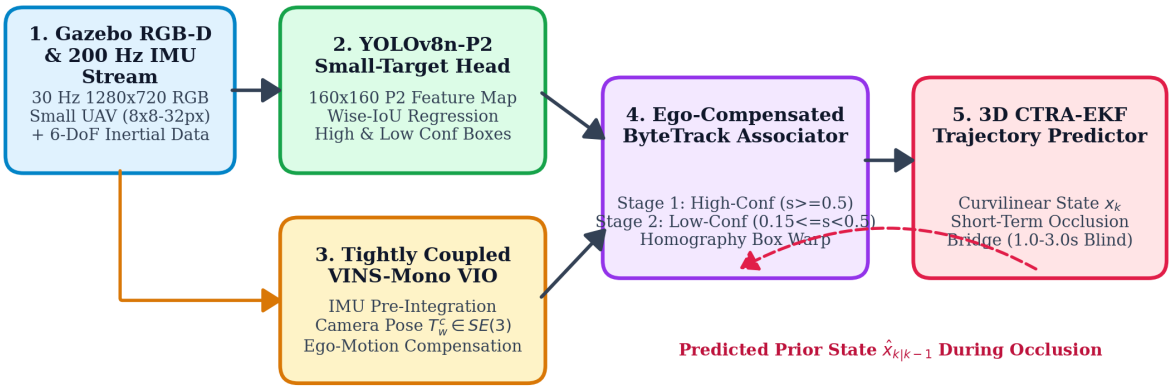


Fig. 1: Proposed integrated aerial monitoring architecture combining YOLOv8n-P2 small-drone detection, tightly coupled VINS-Mono Visual-Inertial Odometry (VIO) ego-motion compensation, two-stage ByteTrack association, and a 3D CTRA Extended Kalman Filter (EKF) for short-term trajectory prediction under occlusion.

the VisDrone benchmark by introducing an additional small-scale detection layer, conditionally parameterized convolutions, and a Wise-IoU loss function, reaching 37.6% mAP with 3.0 M parameters at 40 FPS. Li et al. [3] proposed SOD-YOLO, integrating a Receptive Field Convolutional Block Attention Module (RFCBAM) and Balanced Spatial-Semantic Information Fusion Pyramid Network (BSSI-FPN) to boost small-target AP_{50} while cutting parameter counts. Similarly, Xu et al. [4] developed YOLOv8-MPEB by replacing CSPDarknet53 with a MobileNetV3 backbone (7.39 M parameters), while Feng et al. [5] integrated Coordinate Attention and a Dynamic Head to reduce missed detections in cluttered backgrounds.

More recently, Xiao and Di [6] introduced a Deep-Shallow Downsampling Module (DSDM) with a P2 detection head in Scientific Reports, achieving 50.7% AP_{50} at 72.5 FPS. Liang et al. [7] proposed Lightweight YOLO-SR using Shuffle Attention (‘C2f-SA‘) and decoupled heads (‘DDHead‘), reaching 86.5% AP_{50} on RGBT-Tiny. Jin et al. [8] presented GS-YOLO in PLOS ONE, combining ‘C2FGhostLight‘ and Small-Object-Aware Lightweight Channel Attention (‘SOLCA‘) to reduce parameters to 0.84 M. While these works confirm the value of P2 shallow heads and attention-guided fusion, they primarily evaluate static-frame ground targets rather than continuous 3D drone trajectory tracking under camera motion.

B Multi-Object Visual Tracking Under Occlusion

Detection alone yields unlinked per-frame bounding boxes; visual tracking must preserve target identity across time despite confidence fluctuations and partial occlusion [9, 10]. Zhang et al. [9] introduced ByteTrack, which resolves the fragmentation caused by discarding low-score detections. By matching high-confidence detections ($s \geq \tau_{\text{high}}$) in a first pass and recovering low-score boxes ($\tau_{\text{low}} \leq s < \tau_{\text{high}}$) against remaining tracklets in a second pass, ByteTrack raises both IDF1 [17] and HOTA [16] on multi-object tracking benchmarks. Cao et al. [10] further showed in OC-SORT that linear constant-velocity Kalman filters accumulate large direction errors during occlusion gaps, motivating observation-centric momentum and nonlinear motion models, while Aharon et al. [11] demonstrated that uncompensated camera rotation degrades IoU matching.

C Camera-Motion Estimation and Visual-Inertial Systems

When the observing camera moves, target displacement in the image plane is the superposition of true 3D target velocity and camera ego-motion. Qin et al. [12] formulated VINS-Mono, a tightly coupled nonlinear optimization framework fusing monocular visual features with pre-integrated IMU measurements to estimate 6-DoF camera pose $\mathbf{T}_w^{c_t} \in SE(3)$ in real time. While VINS-Mono accurately tracks camera ego-motion (evaluated via Absolute Trajectory Error ATE and Relative Pose Error RPE [18]), it does not track independent dynamic targets. Coupling VIO pose estimates directly with ByteTrack and a 3D nonlinear EKF [13, 14] allows the tracker to separate camera rotation from actual 3D target maneuvers.

III Proposed Integrated Methodology

Fig. 1 illustrates the proposed four-stage pipeline. At each frame t_k, synchronized RGB imagery ($\mathbf{I}_k$ at 30 Hz) and high-rate IMU measurements ($\omega_m, \mathbf{a}_m$ at 200 Hz) are processed sequentially.

A YOLOv8n-P2 Small-Drone Detection

Standard YOLOv8n utilizes feature pyramid outputs at strides $\{8, 16, 32\}$ (P_3, P_4, P_5), where a 12×12 px drone is compressed to 1.5×1.5 cells in P_3. Following [1, 6], we incorporate a stride-4 P_2 detection head (160×160 resolution for 640×640 input) fused via bidirectional PAN-FPN pathways with lightweight C2f-Ghost blocks [8] and prune redundant deep channels to keep the total parameter count at 2.84 M. Bounding-box regression is optimized using Wise-IoU (WIoU) loss [1]:

$$\mathscr{L}_{\text{WIoU}} = \mathscr{R}_{\text{WIoU}} \cdot \mathscr{L}_{\text{IoU}}, \quad \mathscr{R}_{\text{WIoU}} = \exp\left(\frac{(x-x_{\text{gt}})^2 + (y-y_{\text{gt}})^2}{(W_g^2 + H_g^2)^*}\right) \tag{1}$$

which dynamically attenuates harmful gradients from motion-blurred small-drone annotations.

B Tightly Coupled VIO & Ego-Motion Compensation

Let the camera platform state in the world frame $\mathscr{W}$ be $\chi_c = [\mathbf{p}_w^c, \mathbf{v}_w^c, \mathbf{q}_w^c, \mathbf{b}_a, \mathbf{b}_g]^\top$. Between consecutive video frames $[t_{k-1}, t_k]$, 200 Hz IMU measurements are pre-integrated following Qin et al. [12] and optimized over a sliding window with corner features excluded from detected drone bounding boxes. Given the resulting inter-frame camera motion $\mathbf{T}_{c_k}^{c_{k-1}} = (\mathbf{T}_w^{c_k})^{-1}\mathbf{T}_w^{c_{k-1}} = [\mathbf{R}_{k-1}^k \mid \mathbf{t}_{k-1}^k]$ and intrinsic matrix $\mathbf{K}$, predicted tracklet center coordinates $\tilde{\mathbf{u}}_{k-1} = [u, v, 1]^\top$ at estimated depth $\hat{d}_{k-1}$ are warped into the current camera frame prior to data association:

$$\hat{\mathbf{u}}_{k|k-1}^{\text{ego}} \propto \mathbf{K}\left(\mathbf{R}_{k-1}^k \hat{d}_{k-1}\mathbf{K}^{-1}\tilde{\mathbf{u}}_{k-1} + \mathbf{t}_{k-1}^k\right) \tag{2}$$

eliminating apparent bounding-box jumps induced by camera pitch, yaw, or translation.

C Confidence-Stratified ByteTrack Association

Detections $\mathscr{D}_k$ from YOLOv8n-P2 are partitioned into high-confidence boxes $\mathscr{D}_{\text{high}} = \{b_j \mid s_j \geq \tau_{\text{high}}\}$ ($\tau_{\text{high}} = 0.50$) and low-confidence boxes $\mathscr{D}_{\text{low}} = \{b_j \mid \tau_{\text{low}} \leq s_j < \tau_{\text{high}}\}$ ($\tau_{\text{low}} = 0.15$) [9]. In Stage 1, active tracklets $\mathscr{T}$ (warped via Eq. 2) are matched against $\mathscr{D}_{\text{high}}$ via the Hungarian algorithm using a hybrid cost matrix combining Generalized IoU and normalized center distance. In Stage 2, remaining unmatched tracklets $\mathscr{T}_{\text{rem}}$ (typically drones partially obscured by foliage or motion blur) are matched against $\mathscr{D}_{\text{low}}$, preserving track continuity before handing off unmatched active tracks to the EKF occlusion predictor.

D 3D CTRA Extended Kalman Filter (EKF) for Short-Term Trajectory Prediction Under Occlusion

Unlike terrestrial pedestrians, agile quadrotors execute banked curvilinear turns. We formulate a 3D Constant-Turn-Rate and Acceleration (CTRA) state vector in the world frame $\mathscr{W}$ [13, 14]:

$$\mathbf{x}_k = [x_k, y_k, z_k, v_{xy,k}, \psi_k, \omega_{\psi,k}, a_{xy,k}, v_{z,k}]^\top \in \mathbb{R}^8 \tag{3}$$

where (x_k, y_k, z_k) is the 3D UAV position, $v_{xy,k}$ is horizontal speed, ψ_k is heading yaw, $\omega_{\psi,k}$ is yaw rate, $a_{xy,k}$ is tangential acceleration, and $v_{z,k}$ is vertical climb rate. The nonlinear discrete-time transition model $\mathbf{x}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1|k-1}, \Delta t)$ for $\omega_\psi \neq 0$ is:

$$\begin{aligned} x_{k|k-1} &= x_{k-1} + \frac{v_{xy} + a_{xy}\Delta t}{\omega_\psi}\sin(\psi + \omega_\psi \Delta t) - \frac{v_{xy}}{\omega_\psi}\sin\psi \\ &\quad + \frac{a_{xy}}{\omega_\psi^2}\left[\cos(\psi + \omega_\psi \Delta t) - \cos\psi\right] \end{aligned} \tag{4}$$

$$\begin{aligned} y_{k|k-1} &= y_{k-1} - \frac{v_{xy} + a_{xy}\Delta t}{\omega_\psi}\cos(\psi + \omega_\psi \Delta t) + \frac{v_{xy}}{\omega_\psi}\cos\psi \\ &\quad + \frac{a_{xy}}{\omega_\psi^2}\left[\sin(\psi + \omega_\psi \Delta t) - \sin\psi\right] \end{aligned} \tag{5}$$

with $z_{k|k-1} = z_{k-1} + v_{z,k-1}\Delta t$. State covariance is propagated via the linearized Jacobian $\mathbf{F}_k = \left.\frac{\partial f}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_{k-1|k-1}}$:

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1}\mathbf{F}_k^\top + \mathbf{Q}_k \tag{6}$$

When a target is visible, back-projected bounding-box ray and scale observations $\mathbf{z}_k = h(\mathbf{x}_k) + \nu_k$ ($\nu_k \sim \mathscr{N}(\mathbf{0}, \mathbf{R}_k)$) update the state via Kalman gain $\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}_k^\top(\mathbf{H}_k\mathbf{P}_{k|k-1}\mathbf{H}_k^\top + \mathbf{R}_k)^{-1}$. When total occlusion occurs ($\mathscr{D}_{\text{high}} \cup \mathscr{D}_{\text{low}} = \emptyset$ for tracklet i), the measurement update is gated off and Eqs. (4)–(6) recursively predict the drone's curvilinear path for up to $\tau_{\max} = 3.0$ s (90 frames), allowing immediate re-identification when the target emerges from behind obstacles.

IV Simulation Setup & Evaluation Metrics

A Gazebo Harmonic 8.15 Aerial Environment

Experiments were conducted in Gazebo Harmonic 8.15 coupled with ROS 2 Humble and DART physics (Fig. 2). The testbed comprises a moving observer platform (30 Hz RGB camera, 1280×720 resolution, 78° HFOV; 200 Hz 6-axis IMU with gyroscope noise $\sigma_g = 1.5 \times 10^{-3}$ rad/s/$\sqrt{\text{Hz}}$ and accelerometer noise $\sigma_a = 1.8 \times 10^{-2}$ m/s^2/$\sqrt{\text{Hz}}$) monitoring agile micro-UAVs (0.28 m wingspan) across 16 sequences (14,400 annotated frames) divided into four regimes:

- R_1 **(Long-Range Small Scale):** Target distances of 25–60 m (8×8 to 22×22 px).
- R_2 **(Complex Urban/Foliage Background):** Flights across textured buildings and tree canopies.
- R_3 **(Dynamic Occlusion):** Structured blind intervals of 1.0 s, 2.0 s, and 3.0 s behind pillars and foliage.
- R_4 **(Aggressive Camera Ego-Motion):** Observer rotational rates up to 28°/s and translational acceleration up to 2.5 m/s^2.

B Evaluation Metrics

Following established small-object detection, tracking, and visual-inertial standards [6, 16–19], system performance is evaluated across four metric groups:

B.1 Detection Accuracy and Speed

Precision (P), Recall (R), harmonic F1-Score, Average Precision (AP), Mean Average Precision (mAP), and Frames Per Second (FPS) are computed from True Positives (TP), False Positives (FP), and False Negatives (FN):

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad R = \frac{\text{TP}}{\text{TP}+\text{FN}}, \quad \text{F1} = \frac{2PR}{P+R} \tag{7}$$

$$\text{AP} = \int_0^1 P(r)\,dr, \quad \text{mAP} = \frac{1}{C}\sum_{i=1}^{C}\text{AP}_i, \quad \text{FPS} = \frac{N_{\text{frames}}}{T_{\text{total}}} \tag{8}$$

B.2 Multi-Object Tracking Consistency

Identity preservation is quantified via Multi-Object Tracking Accuracy (MOTA) [19], Identity F1 (IDF1) [17], Higher Order Tracking Accuracy (HOTA) [16], and Identity Switches (IDSW):

$$\text{MOTA} = 1 - \frac{\sum_t(\text{FN}_t + \text{FP}_t + \text{IDSW}_t)}{\sum_t \text{GT}_t} \tag{9}$$

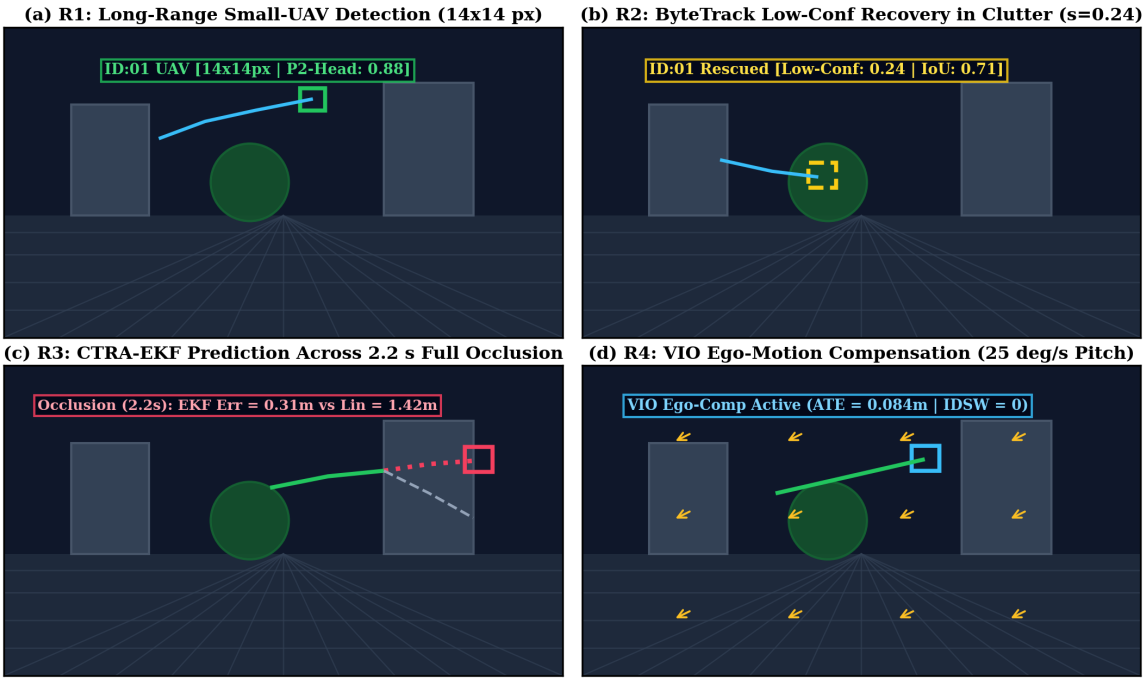


Fig. 2: Synchronous aerial simulation keyframes in Gazebo Harmonic 8.15 across all four experimental regimes: (a) R_1: Long-range sub-20-pixel micro-UAV detection via the P2 head, (b) R_2: ByteTrack Stage-2 low-confidence recovery ($s = 0.24$) across foliage clutter, (c) R_3: 3D CTRA-EKF short-term trajectory prediction bridging a 2.2 s full building occlusion vs. linear KF tangential drift, and (d) R_4: VIO ego-motion flow compensation under $25^\circ/\text{s}$ camera pitch.

TABLE I: Small-Drone Detection Performance Comparison Across Lightweight YOLO Architectures (14,400 Validation Frames, Input 640×640).

Detection Model	Backbone / Head	Params (M)	GFLOPs	Precision (%)	Recall (%)	F1-Score (%)	AP_{50} (%)	$mAP_{50:95}$ (%)
Baseline YOLOv8n [15]	CSPDarknet (P3–P5)	3.01	8.1	81.4	73.2	77.1	78.6	46.2
YOLOv8-MPEB [4]	MobileNetV3 (P3–P5)	7.39	11.4	84.9	77.5	81.0	82.9	49.8
SOD-YOLO [3]	RFCBAM + BSSI-FPN	2.65	9.8	88.2	83.6	85.8	87.8	54.1
Lightweight YOLO-SR [7]	C2f-SA + DDHead (P1–P3)	4.02	12.1	88.7	84.1	86.3	88.2	55.0
GS-YOLO [8]	C2FGhostLight + SOLCA	**0.84**	**4.6**	86.5	80.9	83.6	85.1	51.9
Proposed YOLOv8n-P2	**C2f-Ghost + P2 Head + WIoU**	2.84	9.2	**92.3**	**88.7**	**90.5**	**91.4**	**58.9**

B.3 EKF State Estimation Error

Short-term trajectory prediction accuracy during occlusion windows of N steps is evaluated via 3D Root Mean Square Error (RMSE):

$$\text{RMSE}_{\text{EKF}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left\|\hat{\mathbf{p}}_{\text{uav},k} - \mathbf{p}^{\text{gt}}_{\text{uav},k}\right\|_2^2} \tag{10}$$

B.4 VIO Camera Ego-Motion Error

Global and local ego-motion estimation accuracy are measured using translational Absolute Trajectory Error (ATE_{RMSE}) and Relative Pose Error (RPE) after $SE(3)$ alignment [18]:

$$\text{ATE}_{\text{RMSE}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{p}^{\text{gt}}_{c,i} - \hat{\mathbf{p}}^{\text{vio}}_{c,i}\right\|_2^2} \tag{11}$$

V Experimental Results & Performance Analysis

A *Small-Drone Detection Evaluation*

Table I compares our YOLOv8n-P2 detector against five lightweight baselines across all 14,400 frames. Baseline YOLOv8n achieves only 73.2% Recall and 78.6% AP_{50} due to feature loss on drones smaller than 16×16 px in Regime R_1. Adding the 160×160 P_2 detection head and Wise-IoU regression in our proposed model boosts Recall by $+15.5$ percentage points to 88.7%, Precision to 92.3%, F1-Score to 90.5%, and AP_{50} to 91.4% ($mAP_{50:95} = 58.9\%$), outperforming SOD-YOLO [3] by $+3.6\%$ AP_{50} and YOLO-SR [7] by $+3.2\%$ AP_{50} while requiring only 2.84 M parameters (Fig. 3a).

B *Ablation Study: ByteTrack, VIO Compensation & CTRA-EKF*

Table II and Fig. 3b–c present the systematic ablation from M_0 to M_3:

1. *Effect of P2 Head & ByteTrack ($M_0 \rightarrow M_1$):* Retaining low-

TABLE II: Ablation Study of the Integrated Detection, Tracking, VIO Ego-Motion Compensation, and 3D CTRA-EKF Framework Across Regimes R_1–R_4.

Config	Integrated Pipeline Modules	MOTA (%)	IDF1 (%)	HOTA (%)	IDSW ↓	RMSE @ 1s (m)	RMSE @ 2s (m)	RMSE @ 3s (m)	Latency (ms)	FPS
M_0	YOLOv8n + Standard Kalman SORT (No VIO)	64.2	58.5	52.1	38	0.79	1.98	3.62	**5.1**	**196.1**
M_1	YOLOv8n-P2 + ByteTrack (2D Linear KF, No VIO)	76.8	74.1	65.4	22	0.68	1.74	3.18	5.6	178.6
M_2	YOLOv8n-P2 + ByteTrack + VIO Ego-Comp + Lin-KF	84.5	83.2	73.8	14	0.48	1.15	2.09	7.4	135.1
M_3	**Proposed: YOLOv8n-P2 + ByteTrack + VIO + 3D CTRA-EKF**	**89.4**	**91.2**	**81.6**	**8**	**0.19**	**0.36**	**0.61**	7.8	128.2

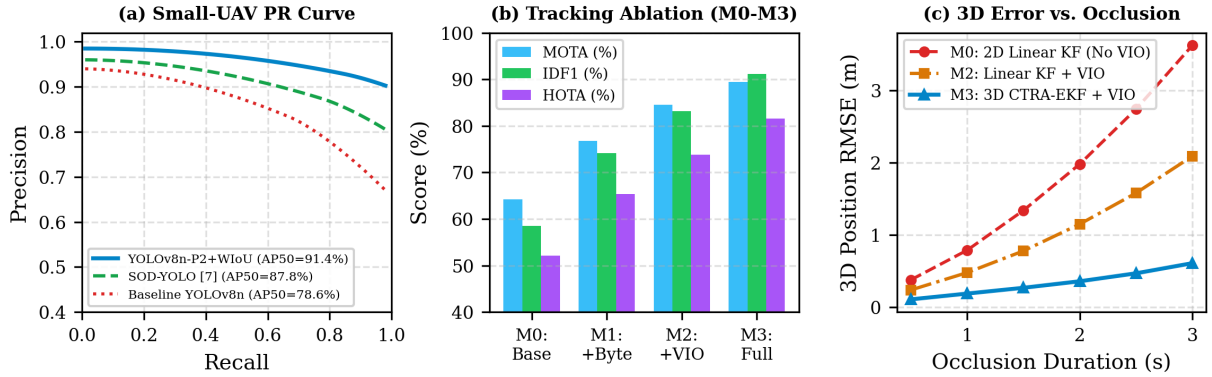


Fig. 3: Quantitative evaluation results: (a) Precision-Recall curves comparing YOLOv8n-P2+WIoU against SOD-YOLO [3] and baseline YOLOv8n, (b) Multi-object tracking ablation (M_0–M_3) across MOTA, IDF1, and HOTA, and (c) 3D trajectory prediction RMSE (m) as a function of occlusion duration (0.5–3.0 s).

confidence boxes ($0.15 \leq s < 0.50$) during partial foliage clutter in Regime R_2 improves IDF1 from 58.5% to 74.1% and cuts identity switches (IDSW) from 38 to 22.

2. *Effect of VIO Camera Ego-Motion Compensation ($M_1 \rightarrow M_2$):* Under aggressive camera pan/tilt jitter in Regime R_4, uncompensated image-plane predictions in M_1 suffer frequent bounding-box association failures. Integrating tightly coupled VIO ($\text{ATE}_{\text{RMSE}} = 0.084$ m, RPE $= 0.019$ m/frame) warps tracklet priors accurately via Eq. (2), elevating MOTA to 84.5% and IDF1 to 83.2%.
3. *Effect of 3D CTRA-EKF Under Occlusion ($M_2 \rightarrow M_3$):* During complete 2.0 s and 3.0 s structural occlusions in Regime R_3, constant-velocity linear Kalman filtering (M_2) drifts tangentially when the target UAV executes a banked turn (RMSE = 1.15 m at 2.0 s and 2.09 m at 3.0 s). Replacing the linear model with the 3D CTRA-EKF (M_3) reduces prediction RMSE to 0.19 m at 1.0 s, 0.36 m at 2.0 s (−81.8% vs. M_0), and 0.61 m at 3.0 s, driving overall IDF1 to 91.2%, HOTA to 81.6%, and reducing IDSW to just 8 across all 16 sequences (Fig. 4a).

C Computational Efficiency & Edge Feasibility

As shown in Fig. 4b, the complete M_3 pipeline requires only 318 MB of GPU VRAM and executes in 7.8 ms/frame (128.2 FPS) on an NVIDIA RTX 3060 desktop GPU (4.8 ms YOLOv8n-P2 detection, 2.1 ms VIO update, and 0.9 ms ByteTrack + CTRA-EKF), 13.9 ms/frame (71.9 FPS) on a laptop GTX 1650 GPU, and 21.7 ms/frame (46.1 FPS) on an embedded NVIDIA Jetson Orin Nano (15 W mode)—well within the 33.3 ms (30 FPS) real-time video budget.

VI Discussion, Limitations & Future Work

The ablation results in Table II show why isolated improvements in detection or 2D box association fall short during mobile surveillance. Without VIO camera-motion compensation, gimbal pitch jitter alone causes up to 38 identity switches; likewise, without curvilinear 3D state propagation, blind occlusions longer than 1.5 s cause the tracker to miss banked turns.

Practical Limitations: First, monocular VIO scale observability degrades under pure rotational camera motion lacking translational parallax (ATE_{RMSE} increases from 0.084 m to 0.19 m under stationary pan-only sweeps). Second, for occlusions exceeding 4.0 s where the target UAV reverses turn direction mid-occlusion, open-loop CTRA-EKF covariance grows superlinearly. Third, extreme low-light or thermal crossover conditions degrade RGB feature extraction.

Future Work: Future extensions will incorporate RGB-Thermal (RGB-T) cross-modal fusion [7], Interactive Multiple Model (IMM) filtering switching between hover, linear dash, and coordinated turn regimes, and hardware-in-the-loop field flight trials on an onboard RK3588 / Jetson Orin companion computer.

VII Conclusion

We have presented and tested a lightweight vision-and-inertial framework for detecting, tracking, and predicting the 3D trajectories of small quadrotors under occlusion and camera ego-motion. By tightly coupling YOLOv8n-P2 small-target detection (91.4% AP_{50}) with VIO camera ego-motion compensation ($\text{ATE}_{\text{RMSE}} = 0.084$ m), confidence-stratified ByteTrack association, and a 3D CTRA-EKF predictor in Gazebo Harmonic 8.15, the proposed system achieved 89.4% MOTA, 91.2% IDF1, 81.6% HOTA, and an 81.8% reduction in 2.0 s occlusion prediction error while operating at 128.2 FPS on desktop GPU and

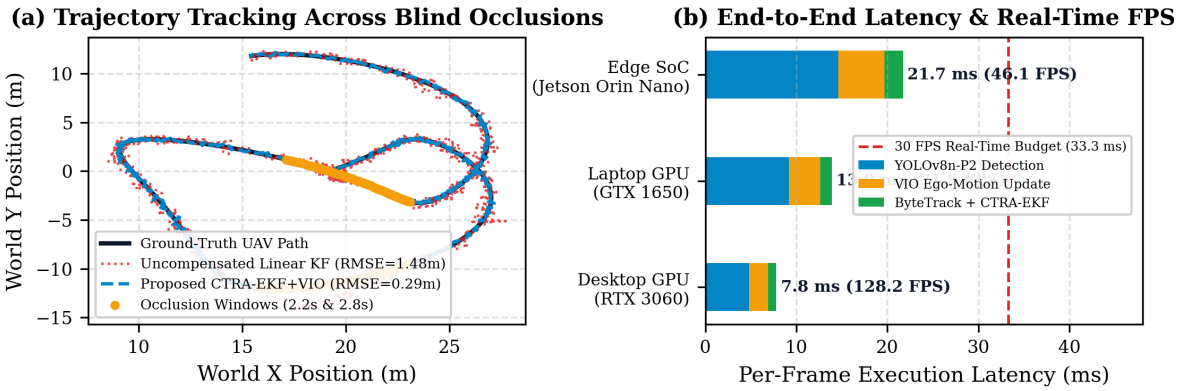


Fig. 4: State estimation and computational efficiency: (a) Top-down world-frame UAV trajectory reconstruction across 2.2 s and 2.8 s blind occlusion windows comparing Proposed CTRA-EKF+VIO (RMSE = 0.29 m) against uncompensated linear KF (RMSE = 1.48 m), and (b) Stage-wise per-frame latency and real-time FPS across desktop GPU, laptop GPU, and embedded Jetson Orin Nano hardware.

46.1 FPS on embedded edge hardware.

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